

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5102

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

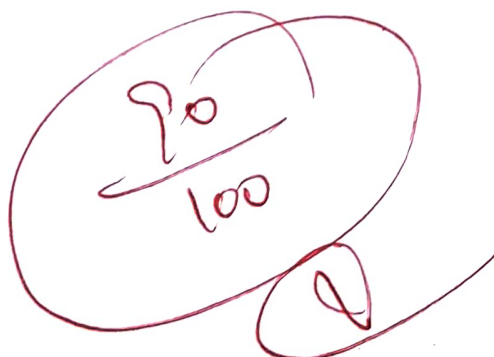
Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

R. Preethi

Signature of the student:

90
100



APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
 - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
 - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
 - a. Deduce the most likely original message.
 - b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
 - a. Demonstrate how the encryption process transforms the message step-by-step.
 - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
 - a. Analyze how investigators could detect the presence of hidden data within the image.
 - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".
 - a. Determine the most probable original message by reversing the encryption process.
 - b. Critically evaluate why such a method is vulnerable in modern communication systems.
 - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

1. Event Communication Security - Caesar CipherScenario

A Students organization wants to send Confidential information about an event. They use a simple classical encryption technique in which every letter is shifted by a fixed number of position in the alphabet.

The original message is:

MEET AT NOON

The shift value is 4 position

a) This is an example of the Caesar cipher

→ In a Caesar cipher each letter of the plaintext is replaced by another letter that is a fixed number of position away from it in the alphabet.

The alphabet position:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X	Y	Z						
16	17	18	19	20	21	22	23	24	25						

$$E(P) = (P + K) \bmod 26$$

$$E(P) = (12 + 4) \bmod 26 \rightarrow (16) \bmod 26 \rightarrow Q$$

$$E(P) = (8 + 4) \bmod 26 \rightarrow 12 \bmod 26 \rightarrow M$$

$$E(P) = (5+4) \bmod 26 \rightarrow 9 \bmod 26 \rightarrow I$$

$$E(P) = (19+4) \bmod 26 \rightarrow 23 \bmod 26 \rightarrow X$$

$$E(P) = (0+4) \bmod 26 \rightarrow 4 \bmod 26 \rightarrow E$$

$$E(P) = (19+4) \bmod 26 \rightarrow 23 \bmod 26 \rightarrow X$$

$$E(P) = (13+4) \bmod 26 \rightarrow 17 \bmod 26 \rightarrow R$$

$$E(P) = (14+4) \bmod 26 \rightarrow 18 \bmod 26 \rightarrow S$$

$$E(P) = (14+4) \bmod 26 \rightarrow 18 \bmod 26 \rightarrow S$$

$$E(P) = (13+4) \bmod 26 \rightarrow 17 \bmod 26 \rightarrow R$$

Spaces are normally kept unchanged

Encrypted message = QITX EX RSSR

The organization can send QITX EX RSSR instead of MEET AT NOON.

- b) The authorized member must know the encryption rule, which is a shift of 4.

→ To decrypt the message, the member perform the reverse operation.

→ Instead of shifting every character 4 position forward, they shift every character 4 position in backward.

$$D(c) = (c - k) \bmod 26 \quad \text{Cipher text} = QITX \quad EX \quad RSSR$$

$$D(c) = (16-4) \bmod 26 \rightarrow 12 \bmod 26 \rightarrow M$$

$$D(c) = (8-4) \bmod 26 \rightarrow 4 \bmod 26 \rightarrow E$$

$$D(c) = (8-4) \bmod 26 \rightarrow 4 \bmod 26 \rightarrow E$$

$$D(c) = (23 - 4) \bmod 26 \rightarrow 19 \bmod 26 \rightarrow T$$

$$D(c) = (4 - 4) \bmod 26 \rightarrow 0 \bmod 26 \rightarrow A$$

$$D(c) = (23 - 4) \bmod 26 \rightarrow 19 \bmod 26 \rightarrow T$$

$$D(c) = (17 - 4) \bmod 26 \rightarrow 13 \bmod 26 \rightarrow N$$

$$D(c) = (18 - 4) \bmod 26 \rightarrow 14 \bmod 26 \rightarrow O$$

$$D(c) = (18 - 4) \bmod 26 \rightarrow 14 \bmod 26 \rightarrow O$$

$$D(c) = (17 - 4) \bmod 26 \rightarrow 13 \bmod 26 \rightarrow N$$

The decryption process can be represented as:

Ciphertext $\rightarrow$ Shift 4 position backward $\rightarrow$ plaintext

Thus, the authorized member can recover:

MEET AT NOON

$\rightarrow$ The important thing is that both sender and receiver must know the secret shift value. In this case, the encryption key is 4.

Conclusion:

The Caesar cipher is simple and easy to implement, but it is not secure for modern communication because there are only 26 possible shifts. An attacker can try all possible shifts and easily discover the original message.

3. Secure messaging system - Vigenere cipher

Scenario

A company uses a keyword-based encryption technique for internal communication.

An employee wants to Send:

HELLO

The keyword is: KEY

- a) The Vigenere cipher uses a keyword to determine different shifts for different letters.

First, assign numbers to the alphabet:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P	Q	R	S	T	U	V	W	X	Y	Z				
15	16	17	18	19	20	21	22	23	24	25				

Plaintext: Hello

Keyword: key

Since the keyword has only 3 letters but the message has 5 letters, the keyword is repeated.

$$E(P) = (P + K) \bmod 26$$

Plaintext	H	E	L	L	O
Plaintext Value (P)	7	4	11	11	14
Key	K	E	Y	K	E
Key Value (K)	10	4	24	10	4
Ciphertext Value (E)	17	8	9	21	18
Ciphertext	R	I	J	V	S

Plaintext: Hello

Key: keyke

Ciphertext: RIJVS

The important feature of the Vigenere's cipher is that the same shift is not used for every character.

$$H + K \rightarrow R$$

$$E + E \rightarrow I$$

$$L + V \rightarrow J$$

$$L + K \rightarrow V$$

$$O + E \rightarrow S$$

The two L in the plaintext character in Hello do not produce the same ciphertext:

$$L + V \rightarrow J$$

$$L + K \rightarrow V$$

This happens because the keyword character changes.

→ This makes the Vigenere's cipher stronger than a basic Caesar cipher because different positions can use different shifts.

Conclusion:

The keyword controls the encryption of each other. When the keyword ends, it starts again from the beginning.

plaintext: Hello

Key : key

ciphertext: RITVS

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

90
100

90